Historical Headrooms versus Current-State Scenarios for Sampled Inventory Peaks

A fixed-design comparison of mature empirical quantile rules

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Abstract

Historical scenario revaluation is a mature financial-risk practice, but reanchoring historical inventory changes at today's position need not improve a forecast of a future inventory peak. We compare a historical own-anchor headroom quantile with a current-anchor scenario quantile and three simple references. The response is the maximum absolute source-reported USD net position over the current and next H recorded observations. A seven-cell specification was frozen before target-specific evaluation on one public market-maker series. The common evaluation sample contains 6,625 origins. At H=6, a 720-label window and the 0.95 quantile, historical headrooms reduce mean pinball loss by USD9,374.86, or 2.08%, relative to current-anchor scenarios. This accompanies more strict breaches and larger overruns. The direction reverses at H=24 and at the 0.90 quantile; four fixed geometric-block resampling ranges all cross zero. These are descriptive finite-record results, without a population superiority test or coverage guarantee. The study clarifies the target, mature-label timing, state-revaluation geometry and floating-point equality mechanisms, and supplies bounded replication code and aggregate evidence. It does not estimate cash needs, profits or the causal effect of changing a dealer's risk policy.

1. Financial question and contribution

An inventory capacity forecast must trade the loss from a sampled peak exceeding an issued level against the cost assigned to unused allowance. The choice of historical representation matters even before a forecasting model is fitted. A dealer can pool the extra absolute exposure observed from each historical starting position, or apply historical signed inventory changes to the current position and then calculate a peak. The second construction follows an established scenario-revaluation principle; the first is a simpler empirical summary. They answer the same forecast question under different uses of historical state.

The question here is deliberately narrow: does own-anchor historical headroom pooling improve a prespecified asymmetric forecast score relative to current-anchor revaluation on this recorded series? Both methods use the current absolute inventory as a known lower floor. Their comparison changes historical representation and anchor treatment together. It cannot identify a pure dimension-reduction benefit or an isolated causal inventory-state effect.

The primary finite-record answer favors headroom pooling by 2.08% of the scenario rule's mean loss. That conclusion has material boundaries. The pooling rule has 332 strict breaches against 254 for the scenario rule; the primary direction reverses in two fixed sensitivities and in five of thirteen calendar months. All four descriptive resampling ranges contain zero. The contribution is an auditable application comparison and clarification of construction effects, rather than a new quantile algorithm, a first financial peak-risk application or a general superiority result.

The source records net inventory under an existing business and risk-management policy. An issued level in this study is a passive forecast, not an intervention. If actual quoting, hedging or limits changed, the path could change as well. Lower forecast loss therefore cannot be translated into trading profit, capital savings or the welfare effect of a new risk limit.

2. Closest applications and mature method transfer

Wisniewski, Lindsay and Lindsay (2020), hereafter W20, already study conformal interval estimation for market makers' net positions on this data source, with rolling calibration, several engines and pinball evaluation. Their experimental response is a scalar hourly signed position and their benchmark forecasts the next time epoch. The present response is a multi-record maximum absolute position including the issue record. Published W20 scores are consequently not directly comparable benchmark numbers. Financial conformal forecasting and use of this source do not originate here.

Peak-risk quantiles also have financial precedents. Van den Hoogen (2017) estimates a maximum cumulative payment-cash deficit using transaction data; Liao and Theodosopoulos (2005) study maximum collateral-adjusted contingent exposure and marking decisions. Payment liquidity, collateral exposure and netted inventory are distinct objects. These sources rule out a broad first-financial-peak claim while not supplying interchangeable empirical benchmarks.

Barone-Adesi, Giannopoulos and Vosper (1999) provide a mature scenario-revaluation precedent. The March 1999 working-paper body consulted uses filtered returns, historical standardized innovation vectors, recursive volatility and price propagation, and portfolio revaluation. Our scenario rule uses complete observed inventory-increment envelopes. It does not estimate a volatility filter, draw fitted innovations, propagate fitted dynamics or price derivatives. Only the current-state scenario-revaluation principle is transferred; filtered historical simulation and its assumptions are not implemented or imported as guarantees.

Quantile scoring follows the established expected-score perspective of Gneiting (2011). Nonexchangeable conformal theory in Barber et al. (2023) and recent financial calibration in Schmitt (2026) supply context, not a theorem for the rules evaluated here. Schmitt's version 3, revised 3 August 2026, is a relevant recent preprint; recency is not treated as evidence of popularity. The geometric-block mechanism follows Politis and Romano (1994), whose inferential premises are separated from the descriptive mechanism used below.

3. Recorded peak target and completed information

Let N_t be the signed source-reported USD net position at zero-based recorded index t. H denotes subsequent recorded observations. Define the known current floor B_t, future sampled peak $P_{t,H}$ and extra headroom $G_{t,H}$ by

$$B_t = |N_t|; \quad P_{t,H} = \max_{0\le h\le H}|N_{t+h}|; \quad G_{t,H} = P_{t,H} - B_t.$$

The maximum includes the current record, so $P_{t,H} \ge B_t$ and $G_{t,H} \ge 0$. An interior spike counts even if the endpoint returns to its starting level. This sampled maximum does not bound an unobserved continuous-time maximum.

For a completed historical origin i retain the directional increment extrema and absolute increment radius:

$$a_{i,H} = \min_{0\le h\le H}(N_{i+h} - N_i); \quad b_{i,H} = \max_{0\le h\le H}(N_{i+h} - N_i).$$

$$D_{i,H} = \max(b_{i,H}, -a_{i,H}); \quad a_{i,H} \le 0 \le b_{i,H}.$$

At actual issue index t the exact calibration pool is $I_t = \{t-H-n+1, \ldots, t-H\}$. It contains n fully completed labels, each ending at $i+H \le t$. The most recent label may end at the already observed issue record. Actual issue time is checked independently against the supplied pool; a later or incomplete label is rejected. The retrospective information convention treats record t as available immediately before issue. The file's historical release and vintage convention is unverified, so this is not evidence of an executable live information set.

Write $Q_k(z)$ for the kth smallest of n entries, keeping duplicates, with $k = \text{ceil}((n+1)\tau)$. The exact rational τ determines the integer rank. No interpolation, jitter or additional decimal rounding is used. An unavailable generic rank is positive infinity; an unavailable configured rank stops the run. The configured ranks below are all available. This corrected prediction rank is not generically the unique empirical pinball minimizer.

4. Five empirical capacity rules

The five methods have fixed reporting order F, S, R, L and P0. Own-anchor pooling F uses historical extra headrooms. Current-anchor scenarios S revalue each completed increment envelope at N_t. The remaining methods are an increment-radius reference, a floored historical peak-level quantile and the current-floor reference:

$$\text{F:} \quad c_F(t) = Q_k(\{G_{i,H}: i \text{ in } I_t\}); \quad C_F(t) = B_t + c_F(t).$$

$$s_i(t) = \max(N_t + b_{i,H}, -N_t - a_{i,H}) - B_t.$$

$$\text{S:} \quad c_S(t) = Q_k(\{s_i(t): i \text{ in } I_t\}); \quad C_S(t) = B_t + c_S(t).$$

$$\text{R:} \quad C_R(t) = B_t + Q_k(\{D_{i,H}: i \text{ in } I_t\}).$$

$$\text{L:} \quad C_L(t) = \max(B_t, Q_k(\{P_{i,H}: i \text{ in } I_t\})); \quad C_{P0}(t) = B_t.$$

Directional extrema are sufficient for this target because the maximum of $|N_t+\text{increment}|$ over an observed path occurs at an increment extremum. S is therefore exact for the revalued historical sampled envelope; it does not retain only the two path endpoints or fit a high-dimensional path model. F also uses the current state, through B_t.

In exact arithmetic both historical and revalued headrooms lie between zero and $D_{i,H}$. Order-statistic monotonicity gives $C_F \le C_R$ and $C_S \le C_R$, and all capacities are at least B_t. There is no universal F/S order or general order between L and the other historical rules. Elementary derivations and coherent counterexamples appear in Appendix A. Geometric upper bounds are not future coverage or financial adequacy guarantees.

Stored float64 arithmetic is part of the declared calculation. S ranks headrooms and then adds B, matching F's order of operations. The private selected c_F/c_S values differ from the public issued extras C_F-B and C_S-B when addition rounds. Four categories distinguish equal complete paired candidate arrays; changed arrays with equal selected allowances; changed selected allowances with equal stored capacities; and different stored capacities. These are computed equality categories, not measures of economic equivalence.

5. Source, frozen design and score

The pinned source is version 2 of davidlindsay1979/net-position-time-series-dataset, using the one-hour member named in the replication README. The accepted CSV has 1,286,391 bytes and 18,576 observations from 5 February 2014 01:00 to 24 January 2017 00:00, in a naive, unverified timezone. Only datetime and NetPosUsd enter the analysis. W20 describes aggregate client-flow positions in five liquid currency pairs. A netted USD scalar cannot reconstruct gross currency exposure, collateral, settlement cash or profit.

There are 18,420 one-hour and 155 forty-nine-hour adjacent gaps. Records are neither sorted nor dropped, and missing calendar intervals are not filled or interpolated. Thus H counts recorded steps, not clock hours. A future path crossing a long gap remains in the primary evaluation.

Six hours of active primary-source reading and scientific synthesis were completed before two hours of active design. The specification was frozen at 2026-10-21 05:46:37 UTC and published in repository commit 6dd404d6e700ff031793d68be209e8f783b22582 before this study's target-specific labels, forecasts or losses were computed. Scheduling gaps and overlapping work were excluded from those active durations. This is a retrospective specification holdout with completed-label feedback, not external preregistration, an external validation dataset or a prospective test. Prior source-paper knowledge and source selection remain limitations.

The primary cell is H=6, n=720, τ=19/20, rank 685. Six one-factor sensitivities use H=12 or 24; n=480 or 1440; and τ=9/10 or 99/100, changing only the indicated primary parameter. They have ranks 685, 685, 457, 1369, 649, 714 respectively. This is seven fixed cells, not a Cartesian search. Every cell uses the same origins: naive timestamp at least 1 January 2016, $t \ge 1445$ for full-menu warmup, and $t \le T-1-24$. The accepted common sample is indices 11927 through 18551 inclusive, from 1 January 2016 00:00 through 23 January 2017 00:00, with M=6,625. Earlier evaluation labels enter later pools only after completion.

Stored error is calculated directly as $e_{t,j} = P_{t,H}-C_j(t)$, and the score is

$$\rho_\tau(e) = \tau \max(e,0) + (1-\tau)\max(-e,0); \quad d_t = \rho_\tau(e_{t,F}) - \rho_\tau(e_{t,S}).$$

The two score weights are converted separately from exact fractions. At τ=0.95 underprediction has nineteen times the weight of overprediction. These are declared forecast preferences, not estimated dealer costs. Under appropriate finite expected-score conditions pinball targets a quantile, but this does not prove conditional optimality of either rolling rule. The primary contrast is the untrimmed equal-origin mean of all d_t, retaining zeros; negative values favor F within that cell. A nonnegative primary mean would fail the specified finite-record improvement criterion. No population test is attached to it.

Additional metrics are mean issued capacity and extra allowance, strict breach count/rate, mean positive overrun, conditional overrun with its breach denominator, and mean unused allowance. A strict breach means $e>0$. Conditional severity with no strict breach is undefined with an explicit reason, not zero. Ordinary means use a compensated finite sum; an exactly constant stored vector retains its constant. Required nonfinite values stop the run.

6. Dependence, diagnostics and descriptive stability

Overlapping peaks, rolling windows and persistent inventory induce dependence. No exchangeability, stationarity, mixing rate or inferential-bootstrap premise has been established for the actual loss sequence. Reanchoring at current inventory does not create conditional exchangeability of historical and future scores. The corrected rank alone supplies no coverage theorem here.

Primary diagnostic panels use current sign, pre-cutoff absolute-inventory terciles, original t modulo6, every inclusive naive calendar month, and whether all H subsequent adjacent gaps equal one hour. The last panel uses retrospective outcome-path information and is neither a predictor nor an inclusion filter. Tercile cut points use ordinary ranks ceil(m/3) and ceil(2m/3), retaining tied or empty groups. Every intentional group retains sixty method-metric rows and one paired-mean row. Groups below 30 origins retain counts/statuses but withhold all metric values uniformly. Display suppression never changes calibration, global scoring or resampling; it is not a minimum-breach rule.

Raw error signs remain unchanged. Auxiliary breaches use $e>32\varepsilon \max(1,P,\text{all five capacities})$, where ε is float64 machine precision; separate near-event and near-F/S-capacity flags are retained. Exact current-floor and maturity violations stop without clipping. Fixed upper-geometry tolerances are arithmetic checks, not permission to remove realized future breaches.

The complete ordered primary contrast is resampled by circular geometric blocks, with fixed expected lengths $\ell=120,240,720,1440$ and 999 means per length. Each length uses PCG64 with SeedSequence([20261020,ℓ]), one continuous stream. A block draws a uniform starting index then a positive geometric length with probability $1/\ell$. It continues circularly and only the final used block is truncated, until exactly M positions have been consumed. Long blocks are not rejected, forecasts are not refitted, and d is not centered on a null.

The observed mean is distinct from the replicate average. Replicate SD is centered on that average with divisor 998; range endpoints are ordinary one-based order statistics 25 and 975, without interpolation or widening. We also report ℓ/M and expected used blocks $1+(M-1)/\ell$; the latter is not an effective sample size. An exactly constant d would use analytic identical replicates with flags; mean zero alone does not trigger that branch. In the actual run neither constant flag is true.

These ranges describe sensitivity of the empirical paired mean to the declared resampling mechanism. They are not population confidence intervals, superiority probabilities or certified standard errors. Executing the Politis–Romano mechanism does not establish its inferential assumptions for this record. A range crossing zero changes neither the observed finite-record mean nor the estimand.

7. Results: loss improvement and risk tradeoffs

Table 1. Primary results. All monetary columns are source-reported USD millions; breach is a percentage. Conditional overrun uses only strict breaches. Full-precision USD values remain in summary.csv.

Method	Loss	Capacity	Extra	Breach %	Overrun	Cond. overrun	Unused
F	0.441774	22.645674	7.413933	5.011	0.166531	3.323097	5.671391
S	0.451149	23.829232	8.597491	3.834	0.116728	3.044580	6.805145
R	0.527544	26.400452	11.168710	1.842	0.064562	3.505941	9.324199
L	0.925885	31.731323	16.499582	7.381	0.196359	2.660285	14.786868
P0	1.813620	15.231741	0.000000	77.011	1.909073	2.478952	0.000000

F has the lowest primary mean pinball score, USD0.441774 million versus S's USD0.451149 million. The unrounded paired mean is −USD9,374.860250076379, or −0.009374860250076379 million; relative to S loss it is −0.02077996933256255. Monetary tables use a fixed 1e−6 display factor. Rounded displayed method means are not used to reconstruct the paired contrast.

The loss improvement is accompanied by lower mean capacity, extra allowance and unused allowance, but more strict breaches: 332/6625 for F versus 254/6625 for S. F's unconditional positive overrun is USD0.166531 million

against 0.116728 million, and conditional overrun is 3.323097 million against 3.044580 million. It is therefore a score tradeoff, not safety dominance.

R has the fewest primary breaches (122) and lowest mean positive overrun, but larger capacity and pinball loss. Its conditional severity,3.505941 million over 122 breaches, exceeds F/S. L has the greatest mean capacity yet 489 breaches, showing why aggregate mean capacity cannot establish pathwise dominance. P0 has zero extra and unused allowance, but 5102 breaches and the largest loss. Its lower conditional severity,2.478952 million over 5102 events, must be read with that different denominator.

Table 2. Event denominators and numerical boundary counts for the primary cell. Strict/auxiliary breach disagreements are zero for all five methods.

Method	Strict breaches	Conditional denominator	Near-event count
F	332	332	0
S	254	254	0
R	122	122	0
L	489	489	95
P0	5102	5102	1523

F capacity is lower than S at 5509 origins, equal at 809, and higher at 307. The mechanism partition is 0 equal paired candidate arrays, 558 changed arrays with equal selected allowances, 251 different selected allowances collapsing to equal stored capacities, and 5816 different capacities. The 251 rounding cases must not be described as equality of selected allowances. The near-F/S capacity count is 809; categories themselves use exact computed equality, not a tolerance. These results do not isolate a causal state effect.

8. Prespecified sensitivities and diagnostics

Table 3. Complete fixed menu: mean loss for every method and F-minus-S paired mean d. Monetary values are USD millions. Rows are in frozen order; different τ rows express different preferences.

Cell	H	n	τ	F	S	R	L	P0	d
main	6	720	0.95	0.441774	0.451149	0.527544	0.925885	1.813620	-0.009375
h12	12	720	0.95	0.592518	0.604781	0.689604	1.058614	2.950594	-0.012263
h24	24	720	0.95	0.777069	0.748384	0.803106	1.197833	4.630827	+0.028685
n480	6	480	0.95	0.449843	0.460816	0.532515	0.885243	1.813620	-0.010973
n1440	6	1440	0.95	0.443793	0.448275	0.526979	0.931562	1.813620	-0.004482
tau90	6	720	0.90	0.659643	0.659499	0.763481	1.477201	1.718166	+0.000144
tau99	6	720	0.99	0.154991	0.158998	0.175486	0.288834	1.889983	-0.004007

The primary direction persists at H=12 and at windows 480 and 1440, and at τ=0.99. It reverses at H=24, with paired mean +USD28,685.304974 (+3.833% of S), and at τ=0.90, with +USD144.225522 (+0.02187%). The latter small positive is retained rather than rounded to zero. Five negative and two positive cells are correlated results from the fixed menu, not independent replications or a post-result majority-vote criterion. Figure 1 displays all seven within-cell contrasts. The primary cell remains unchanged.

Table 4. All primary calendar-month contrasts, in USD millions, with strict breach counts. Calendar groups use the naive source timestamps.

Month	Origins	F−S mean	F breaches	S breaches
2016-01	504	-0.017183	29	18
2016-02	504	-0.015726	21	18
2016-03	552	+0.040380	42	31
2016-04	502	+0.020270	37	24
2016-05	530	-0.031114	10	7
2016-06	528	-0.035273	16	7
2016-07	504	+0.006976	34	27
2016-08	552	-0.047937	14	12
2016-09	526	+0.002870	31	29
2016-10	506	-0.007725	39	36
2016-11	528	+0.007843	25	24
2016-12	526	-0.032346	15	11
2017-01	363	-0.012823	19	10

Eight monthly means favor F and five favor S. March 2016 has the largest adverse displayed value (+USD40,379.69), while August 2016 is most negative (−USD47,937.36). F has more strict breaches and larger unconditional positive overrun than S in every month, including months where its pinball loss is lower. Figure 2 retains all thirteen months. None is an independent episode or a proposed switching regime.

Table 5. Complete remaining primary paired-mean panels. Monetary values are USD millions. Undefined means correspond to an intentional empty group.

Panel	Group	Origins	F−S mean
Sign	negative	5592	-0.010808
Sign	zero	0	undefined
Sign	positive	1033	-0.001618
Dev. magnitude	low	420	+0.023570
Dev. magnitude	middle	760	+0.016090
Dev. magnitude	high	5445	-0.015470
Index modulo6	0	1104	-0.012793
Index modulo6	1	1104	-0.016473
Index modulo6	2	1104	-0.005393
Index modulo6	3	1104	-0.005242
Index modulo6	4	1104	-0.005957
Index modulo6	5	1105	-0.010390
Future gap	gap_free	6289	-0.006642
Future gap	has_gap	336	-0.060535

Both nonempty sign groups favor F, whereas the low and middle development-magnitude groups favor S and the high group favors F. All six source-index phases have negative paired means; phase partitioning does not remove rolling-window or financial dependence. Both future-gap groups favor F, but the has-gap group has 336 origins and only one strict F breach and one strict S breach. Its conditional severity comparison consequently rests on one event

per method. The zero-current group has no origins and retains 61 null/status rows. There are no below 30 nonempty groups in this run. These predetermined panels qualify the global result without creating new training regimes.

9. All four resampling profiles

Table 6. Descriptive primary resampling profiles. Average, SD and endpoints are USD millions. Every row has 999 replicate identities and observed mean −0.009374860250076379 million.

ℓ	ℓ/M	Expected blocks	Rep. average	Rep. SD	Order25	Order975
120	0.0181	56.2	-0.009774	0.009310	-0.028186	0.008559
240	0.0362	28.6	-0.009835	0.008948	-0.028279	0.008127
720	0.1087	10.2	-0.009471	0.007557	-0.025106	0.004780
1440	0.2174	5.6	-0.009475	0.006679	-0.023037	0.003596

All four actual ranges have negative lower endpoints and positive upper endpoints. The observed mean lies within each. For ℓ=120 the endpoints are −USD28,186.30 and +USD8, 558.86; for ℓ=1440 they are −USD23,036.77 and +USD3,595.62. Replicate SD declines across these four declared lengths, but that empirical pattern is not a basis for selecting a favorable length or claiming general uncertainty monotonicity. Figure 3 shows all endpoints, the separate observed point and replicate averages. The observed primary advantage remains a finite-record score result.

10. Replication evidence and practical limits

Seventy-four artificial tests passed before the final replication. They include independent hand-valued path examples, mature-label and spike checks, exact fractional ranks, score complement weights, chronological alignment, explicit circular-block arithmetic, constant profiles, public scalar/type/null boundaries, corrupted sources, bounded transport and safe failure behavior. Independent reviews checked core, report and orchestration interfaces. Two early attempts stopped before source loading because a macOS thread-table format was not recognized; the format was reproduced, covered by regression fixtures and repaired. Those attempts did not produce financial outcomes.

The published run manifest identifies attempt 20261021T060715Z-de1704c9, the frozen specification hash, the exact source and all eight scientific code/test hashes. All thirteen checks are PASS. The four tables contain 35, 7, 4 and 1647 data rows. Three original vector figures are generated solely from public aggregates. A second complete execution, after replacing account/path text in an artificial operating-system fixture with invented text, produced byte-identical scientific tables and figures; the scientific formulas were unchanged.

The final run recorded 8.02145 seconds from its global start to the final pre-manifest checkpoint, process peak RSS 80,936,960 bytes (77.19MiB), and maximum observed OS thread count 1 across 565 snapshots. The earlier complete run used 6.88405 seconds and82.95MiB. Five numerical thread requests were set to 1 before NumPy import. A snapshot maximum is not a continuous thread-peak observation; the reported resource checkpoint precedes the metadata-write tail. The monitor is macOS-specific and rejects unsupported measurement rather than reporting zero usage. These observed runs remain far below the declared 2GiB and ten-minute budgets.

The dataset was anonymously accessible through a one-file route, historically verified at 02:42 UTC on 21 October 2026. The two analysis runs reported above reused a verified local snapshot and made no download request. The observed rights label is Data files © Original Authors, and raw redistribution permission is unverified. The repository therefore distributes acquisition code, member/version/hash information and aggregate evidence, excluding raw CSV/ZIP files, third-party papers, origin traces, private development thresholds and complete replicates.

A separate missing-cache acquisition and reproduction check at 06:24 UTC on 21 October 2026 exercised the published anonymous one-file route. It accepted the same CSV identity and produced byte-identical copies of all four scientific tables and three figures, with unchanged code identities. This additional check recorded 8.91694 seconds, peak RSS 86,654,976 bytes (82.64MiB), and one observed OS thread. Its sanitized source and resource evidence is supplied separately as results/acquisition-verification.json. The primary published run remains the local-reuse attempt identified above; this check changed no scientific specification.

External validity is limited by one old aggregate inventory record, uncertain timezone and publication timing, netting across currencies, sampled rather than continuous extrema, and the business policy that generated the record. No live forecast, causal policy effect or profit was measured. Fixed diagnostics do not establish conditional calibration; few conditional events can give unstable severity. The methods are cheap and transparent, but changing financial laws and dependence remain unresolved. Further validation would require separately specified later records or additional dealers with legitimate access, preserving issue-time information and costs; those data are not part of this study.

11. Conclusion

Applying a mature historical-scenario principle to a sampled inventory-peak target produces a transparent comparison with own-anchor headroom pooling. In the primary fixed setting, pooling has 2.08% lower mean pinball loss, with more breaches and larger overruns. Adverse horizon and preference sensitivities, monthly and magnitude-group reversals, and four resampling ranges crossing zero prevent a general superiority conclusion. The useful finding is the limited and reproducible score tradeoff, together with the distinction between known-floor geometry, state revaluation, computed equality and unproved coverage. The evidence supports further targeted validation, rather than a new risk-control guarantee.

Appendix A. Elementary geometry and coherent examples

These observations are direct consequences of maxima, absolute values and order statistics, not new theoretical results. First, if $C<B\le P$ and $0<\tau<1$, both C and B underpredict P and $\rho_\tau(P-C)-\rho_\tau(P-B)=\tau(B-C)>0$. Raising a subfloor capacity to B strictly improves the real-arithmetic score. Since all five stored rules obey $C\ge B$, strict breach indicators and raw overruns cannot exceed P0 at the same origin. This mechanical bound orders neither pinball loss nor conditional severity. Exact mathematical strictness does not require distinct rounded stored losses.

For one fixed completed increment envelope write $u=b\ge 0$, $v=-a\ge 0$, $l=\min(u,v)$, $r=\max(u,v)$, and $g(x)=\max(x+u,-x+v)-|x|$. Splitting at $x=0$ shows $0\le l\le g(x)\le r$. More explicitly, when $u>v$, $g(x)=v+\mathrm{clip}(u-v+2x,0,u-v)$; when $v>u$, $g(x)=u+\mathrm{clip}(v-u-2x,0,v-u)$; when $u=v$, $g(x)=u$. Therefore historical own-anchor and current-anchor scores both lie between the same paired l_i and r_i. Monotonicity gives the fixed-pool bound

$$Q_k(l)\le c_F,\ c_S\le Q_k(r);\quad |c_F-c_S|\le Q_k(r)-Q_k(l).$$

This is a bound on fixed historical scores, not on a future distribution, full forecasting performance, a causal effect or general convexity. The same reasoning with r=D gives the $F/S\le R$ capacity relations.

Table A1. Artificial $H=1,n=3,\tau=3/4$ examples at issue t=3 (rank3). Units are invented. Both complete histories and subsequent peaks are coherent.

History N0…N3	CF	CS	CR	CL	CP0	Next N4	LF−LS
6,4,2,0	0	2	2	6	0	1	+0.50
6,4,2,0	0	2	2	6	0	0	−0.50
0,−4,−1,2	6	5	6	4	2	7	−0.75
0,−4,−1,2	6	5	6	4	2	2	+0.25

The first history has smaller F capacity but either sign of realized loss difference, depending on the next observation. The second history has larger F capacity and again either sign. Neither an F/S capacity order nor a score order follows universally. These invented examples verify interpretation and were not used to select a favorable empirical specification.

Finally, the target has a known floor and can have an atom there. For a fixed conditional target distribution, a quantile C satisfies $F(C-)\le\tau\le F(C)$; strict exceedance need not equal $1-\tau$. An empirical breach rate near 5% is consequently neither a proof of calibration nor a reason to replace the score criterion. This statement about ideal quantiles does not assume that either historical rule estimates the true conditional distribution.

Appendix B. Output and numerical conventions

The machine-readable results/run-manifest.json records the frozen scientific specification and its schema binding, the ordered common-origin fingerprint, the accepted source identity and exact code identities. Each origin in that fingerprint is encoded as an unsigned eight-byte little-endian integer. The root README gives the full accepted CSV SHA-256 and member/version. Full artifact paths, byte counts and SHA-256 values are retained in the manifest, allowing an independent check without redistributing the source observations.

The public tables contain builtin finite scalar leaves only. CSV empty values represent documented undefined quantities, not numerical zeros; JSON forbids nonfinite values. Mean loss reconciliation and weighted diagnostic reconciliation use the declared arithmetic tolerance, separately from event flags. Constant-input resampling preserves exact stored constants. The production bootstrap uses at most two float64 slice sums per circular block, combining segment totals with a compensated sum, without a replicate-by-origin matrix or unstable global-prefix subtraction.

Source validation hashes and parses the same immutable opened-file snapshot. Explicit acquisition is missing-source-only, anonymous and capped at 25MiB, with a fixed single allowed302, no implicit proxy, no reading of redirect/error bodies, exact ZIP member/CRC/size/hash checks and exclusive acceptance into a task cache. Existing wrong inputs are never overwritten. Output attempts are exclusive, partial completed cells/lengths are retained privately, and a sanitized manifest is written last. COMPLETE identifies the scientific scope, not manuscript delivery or external validation.

References

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Figure captions

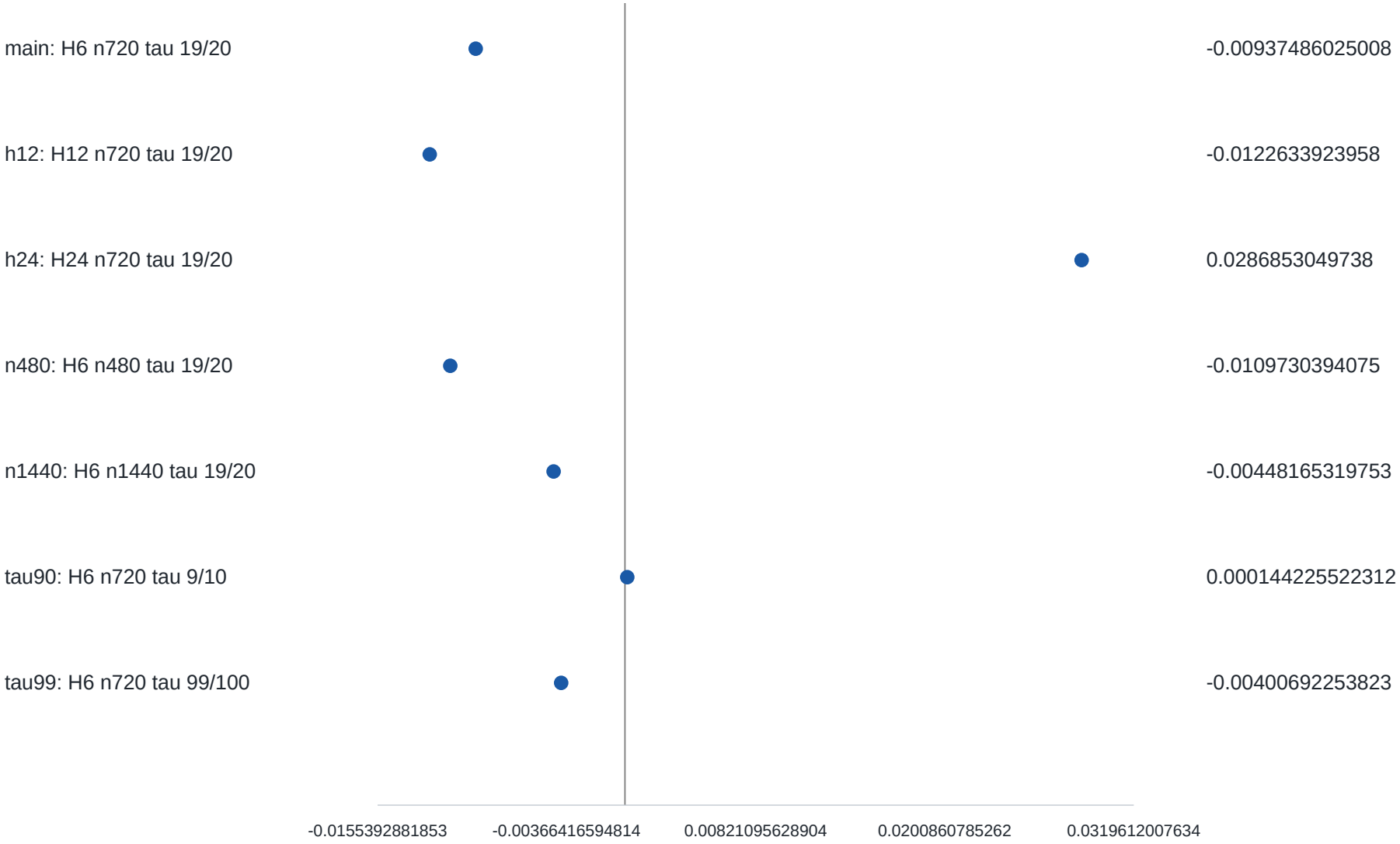
Figure 1. All seven within-cell paired means. Negative values favor F for that fixed objective; different preferences are not pooled into an across-cell ranking.

Figure 2. All primary naive-calendar monthly paired means and origin counts. Calendar groups do not constitute independent episodes or a switching strategy.

Figure 3. Four fixed descriptive geometric-block resampling ranges, separate observed mean and replicate averages. These are not confidence intervals or superiority probabilities.

Within-cell paired mean loss differences

F minus S; USD millions. Negative values favor F within the same fixed cell.

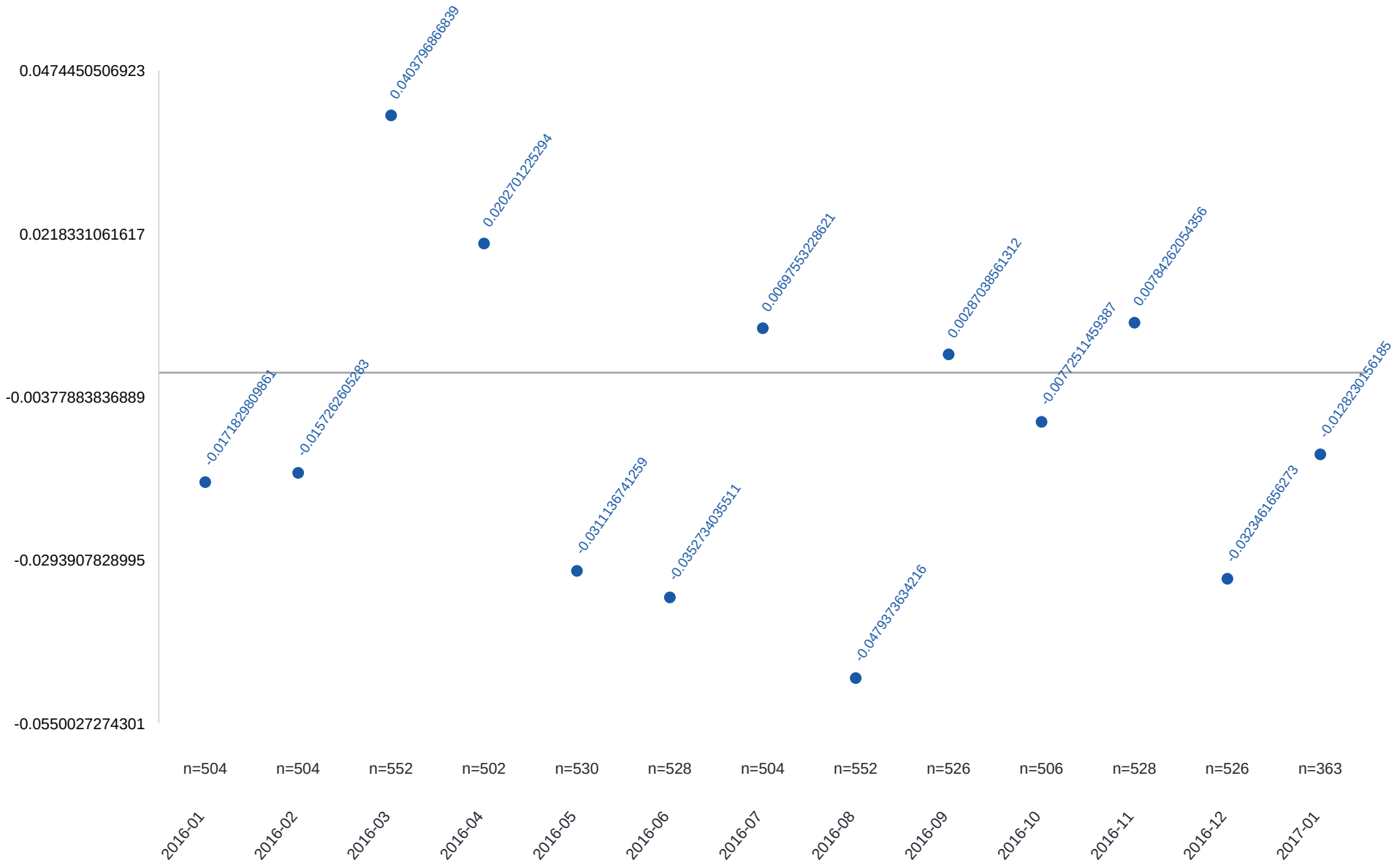


Seven declared cells in fixed order; no comparison of raw losses across different tau objectives.

Points are finite-record paired means; the zero line is descriptive, not an inferential threshold.

Primary monthly paired means and origin counts

F minus S; USD millions. Every inclusive month is retained; points are never joined across gaps.

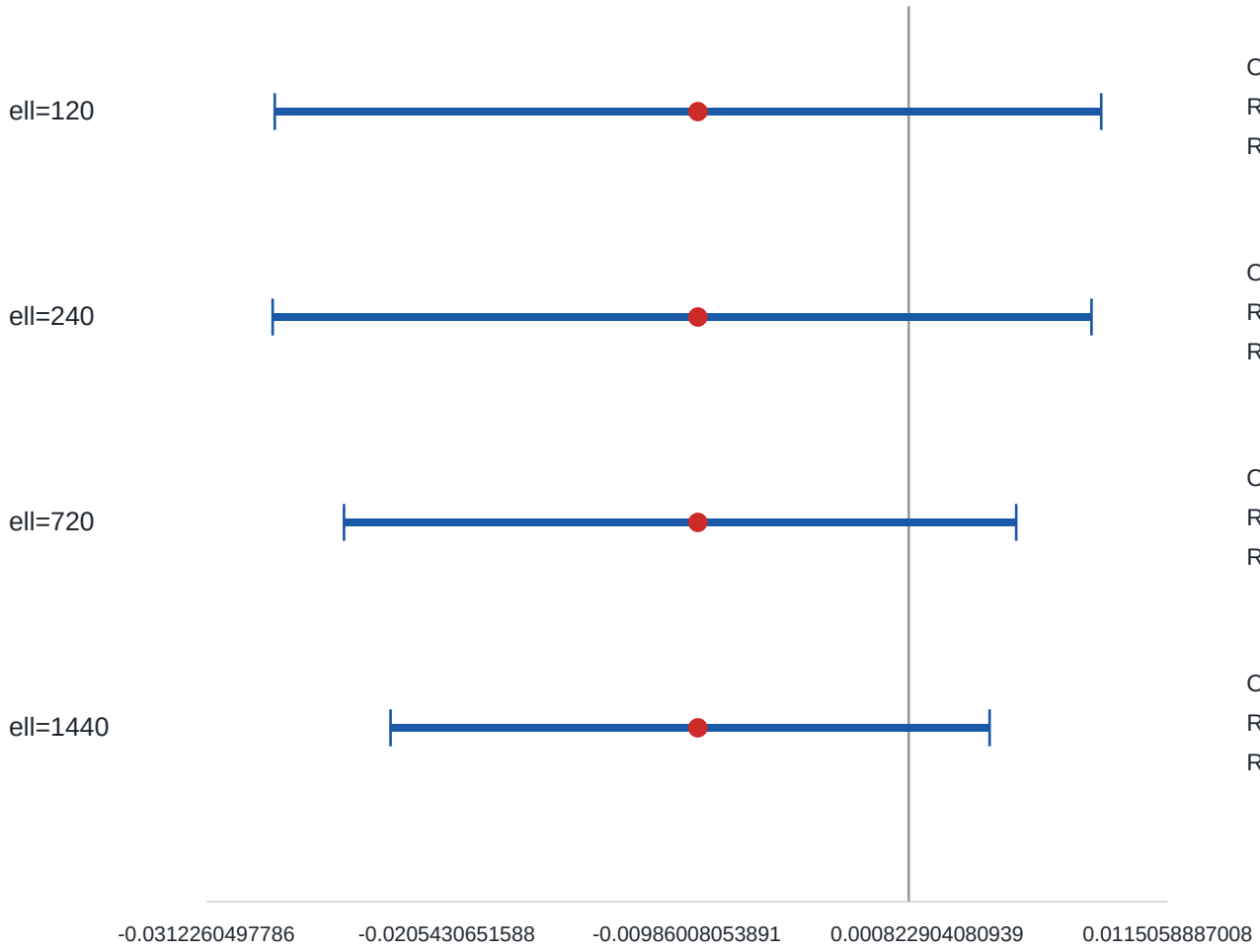


DISPLAYED requires at least 30 origins. Empty/suppressed months have no fabricated numerical mean.

Group display rules do not remove origins from global scoring, calibration or resampling.

Primary descriptive resampling-stability ranges

USD millions. Blue bars: stored order25/order975 endpoints. Red point: separate observed paired mean.



ell=120
Observed: -0.00937486025008
Range: [-0.0281863036743, 0.00855885846083]
Resampled average: -0.00977399659435

ell=240
Observed: -0.00937486025008
Range: [-0.0282790195387, 0.00812681095147]
Resampled average: -0.00983533103288

ell=720
Observed: -0.00937486025008
Range: [-0.0251064775090, 0.00477992887892]
Resampled average: -0.00947100966779

ell=1440
Observed: -0.00937486025008
Range: [-0.0230367747370, 0.00359561739641]
Resampled average: -0.00947488384510

All four fixed lengths, each with 999 replicate identities; no widening, jitter or favorable-length selection.

An observed point may lie outside a resampled range. Constant-input rows use analytic identities.

These are descriptive paired resampling ranges, not population confidence intervals or superiority probabilities.